

RISE Student Research Grant Proposal

Part A: Project Proposal Summary

A.1 Proposed title	
Modular Sparse Autoencoders for Multi-Site Post-Translational Modification Interpretability in Protein Language Models	
A.2 Research area	
Bioinformatics and Computational Biology; Mechanistic Interpretability; Deep Learning for Protein Sequences; Machine Learning Systems	
A.3 Research summary (max. 150 words)	
Protein language models capture rich representations, but individual neurons are polysemantic, meaning a single hidden dimension responds to multiple unrelated concepts. This obscures how models capture post-translational modifications (PTMs) and multi-site crosstalk. While Sparse Autoencoders (SAEs) can unpack overlapping features into interpretable dictionaries, standard monolithic SAEs assume features combine purely additively. In practice, this causes them to collapse into basic amino acid detectors rather than identifying true modification events. We propose a modular SAE framework to isolate modification contexts from sequence backbones. The design explores domain-specialized sub-dictionaries, multi-scale hierarchical grouping, and bilinear interaction terms. We benchmark existing monolithic models (InterProt, InterPLM) against modular designs on pre-extracted PLM representations, evaluating learned features through 2x2 contingency tables, chemistry-stratified permutation tests, and causal activation patching. We will release our modular architecture and dictionaries to support downstream tasks, including multi-label PTM site prediction, missense variant effect analysis, and crosstalk modeling.	
A.4 Project duration (in months)	7
A.5 Total project cost (in BDT)	70,633

Part B: Research Project Details

B.1 Background and statement of the problem (Word limit: Max. 200)
Pretrained protein language models encode detailed structural and evolutionary patterns [1], but their dense representations are heavily entangled. Individual activation dimensions fire for multiple biological concepts at once, making it hard to extract mechanistic insights into post-translational modifications (PTMs) and their combinatorial interactions [2]. In cellular signaling, modifications frequently interact: one modification can recruit or block enzymes at nearby residues, creating complex combinatorial logic gates. Recent studies use Sparse Autoencoders (SAEs) to unpack these representations into sparse dictionaries [3, 4]. However, existing protein SAEs are monolithic and assume that features combine in a simple linear sum [3, 4]. This causes two major failure modes: features often collapse into recognizing common amino acids rather than modifications, and they miss non-linear interactions between adjacent sites. We hypothesize that modular architectures (using domain-specialized sub-dictionaries, multi-scale hierarchical grouping, and bilinear interaction terms) can isolate specific modification states from sequence backgrounds while capturing combinatorial crosstalk without losing reconstruction accuracy [5]. Beyond mechanistic interpretation, these decoupled representations provide useful features for downstream tasks, including multi-label PTM site prediction, missense variant pathogenicity analysis, and combinatorial crosstalk modeling. [1] Z. Lin et al., "Evolutionary-scale prediction of atomic-level protein structure with a language model," <i>Science</i>, 379, 1123-1130 (2023). [2] C. Zhang et al., "COMPASS-PTM: Multi-site post-translational modification profiling at single-residue resolution," <i>Nature Communications</i>, 17, 6450 (2026). [3] E. Simon and J. Zou, "InterPLM: Interpreting protein language models via sparse autoencoders," <i>Nature Methods</i>, 22, 412-422 (2025). [4] L. Adams et al., "InterProt: Scaling sparse autoencoders to foundational protein representations," <i>International</i>

Conference on Machine Learning (ICML 2025).

[5] S. Rajamanoharan et al., "Improving dictionary learning in language models with gated sparse autoencoders," arXiv:2404.16014 (2024).

B.2 Specific objectives (Word limit: Max. 200)

1. To pre-compute and extract intermediate representations from protein language models across reviewed human proteins in Swiss-Prot, harmonizing multi-source experimental modification annotations into strict sequence-identity discovery and held-out evaluation partitions.
2. To explore and test modular and hierarchical designs—including specialized sub-dictionaries, Cascaded SAEs (CSAE), hierarchical structures (HiSAE parent-child gating and Matryoshka representations), and PolySAE bilinear terms—and build a composable modular SAE combining chemistry-partitioned sub-dictionaries with low-rank bilinear interaction heads to resolve multi-site combinatorial modifications.
3. To systematically benchmark baseline monolithic SAE families (InterProt TopK, InterPLM L1-ReLU, JumpReLU) alongside the explored modular architectures to measure reconstruction error, sparsity, dead latent rates, and amino acid collapse, evaluating downstream utility across multi-label PTM site prediction and variant effect benchmarks.
4. To validate learned features using 2x2 contingency tables, 1,000-iteration chemistry-stratified permutation tests, Bonferroni correction, and in silico activation patching to confirm causal biological relevance.

B.3 Expected outcomes (Word limit: Max. 150)

- 1. Curated PTM Benchmark Dataset:** A high-quality evaluation corpus curated from major public repositories (dbPTM, CPLM, qPTM) with extensive filtering to eliminate homology artifacts, annotation inconsistencies, and sequence leakage across identity-clustered splits.
- 2. Composable Modular SAE Architecture:** A documented, open-source PyTorch implementation of our modular SAE architecture, structured to be cleanly composable so future research can integrate custom dictionary heads or extend the design.
- 3. Systematic Comparative Benchmark and Utility Evaluation:** An empirical report cataloging reconstruction accuracy, sparsity, dead latent counts, and utility across multi-label PTM site prediction and variant effect benchmarks.
- 4. Pretrained and Validated Feature Dictionaries:** Open-access modular SAE weights targeting PLM representations, providing statistically and causally validated, interpretable latents across multiple post-translational modification classes.

B.4 Methodology (Word limit: Max. 500)

Data & Preliminary Work: We will curate reviewed human protein sequences from UniProtKB/Swiss-Prot (Release 2025_01). Experimentally verified PTM annotations will be compiled from dbPTM, CPLM 4.0, qPTM, and O-GlcNAcAtlas, strictly excluding computational predictions, homology transfers, and inconsistent records. To prevent data leakage, sequences will be clustered at strict sequence identity using CD-HIT into discovery and held-out evaluation splits, stratified across modifiable residue classes. We will extract representations from frozen ESM-2 Layer 24. We will then evaluate public baselines, specifically InterProt (TopK, k=64, 4,096 dictionary) and InterPLM (L1-ReLU, 10,240 dictionary across 650M and 8M scales), alongside JumpReLU models to establish the baseline performance floor, and verify pipeline feasibility.

Model Architecture & Pipeline: Rather than relying on flat monolithic dictionaries, we will investigate modular Sparse Autoencoder architectures designed to separate modification states from underlying sequence context. We will explore modular candidate components: domain-specialized sub-dictionaries dedicated to specific biochemical classes, Cascaded SAEs (CSAE) that learn higher-level pathway abstractions on decoder weights, hierarchical multi-scale structures (HiSAE parent-child conditional gating and Matryoshka representations), and PolySAE bilinear tensor interactions. Based

on these evaluations, we will build a composable modular SAE that decouples residue identities from functional modification states and captures multi-site combinatorial interactions.

Validation & Leakage Control: All evaluations will be conducted strictly on held-out sequence-identity clusters to guarantee generalizability. Latent interpretability is quantified via 2x2 contingency tables across individual PTM classes, computing fold enrichment, selectivity odds ratios, and rank correlation metrics. To verify that features detect authentic biological modifications rather than background amino acid prevalence, we will execute chemistry-stratified permutation null tests shuffled within residue strata, applying Bonferroni family-wise error correction. Dictionaries are screened through strict quality gates requiring high explained variance, low dead-latent frequencies, and exclusion of latents exhibiting amino-acid dominance.

Challenges & Mitigation:

- 1. Compute and Runtime Reliability:** Full-proteome streaming passes require continuous GPU execution that risks session timeouts on free platforms. *Mitigation:* We will use streaming histogram accumulation, discarding raw representations after each mini-batch to keep active memory minimal, and use Google Colab Pro+ for background execution and reliable GPU allocation.
- 2. Amino Acid Feature Collapse:** Unsupervised autoencoders often learn trivial amino acid detectors. *Mitigation:* We will partition dictionary heads by residue chemistry and enforce within-stratum permutation nulls.
- 3. Multiple Testing False Positives:** Testing thousands of features risks false discoveries. *Mitigation:* We will apply dictionary-level Bonferroni correction combined with empirical permutation percentiles.

Part C: Proposed Grant Amount Summary

(Detail budget must be submitted in a separate file.)

SN	Budget description	Total amount (BDT)
(i)	Resources (Google Colab Pro+ GPU Subscription & LLM Research Tooling)	59,248
(ii)	Miscellaneous (Operations Contingency @ 2%)	1,385
(iii)	Conference Registration and Participation Expenses (maximum BDT 10,000)	10,000
Grand total amount (i+ii+iii)		70,633
(in words) Seventy Thousand Six Hundred Thirty-Three Taka Only.		

Part D: Student Details

D.1 Applicant-1	
Name	Mustafa Muhaimin (Student ID: 2105178)
Department	Computer Science and Engineering, BUET
Email address	2105178@ugrad.cse.buet.ac.bd
Contact no.	01782887798
Research experience (Word limit: Max. 100)	1. Implementation of continuous normalizing flows via flow matching and autoregressive context encoders in PyTorch for high-dimensional sequence generation. 2. Proficient in GPU training optimization, custom loss function design,

	and multimodal embedding conditioning. Competed in the NSUSEC Datathon, building end-to-end feature engineering and automated hyperparameter tuning pipelines for tabular predictive modeling. 3. Academic project work includes developing heuristic search systems, compiler design, and low-level performance benchmarking.
D.2 Applicant-2	
Name	Mesbah Uddin Ahamed (Student ID: 2105139)
Department	Computer Science and Engineering, BUET
Email address	2105139@ugrad.cse.buet.ac.bd
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Research experience (Word limit: Max. 100)	1. Experienced in leakage-controlled evaluation workflows, NLP pipelines for hallucination detection via retrieval-augmented reranking, dialect-to-dialect translation across four Bengali dialects, admission-script OCR verification, and a DDoS prevention capstone for high-volume networks. 2. Responsible for dataset curation, including Bengali dialect audio collection, and executing benchmarking pipelines. Champion, Bengali Long-Form ASR and Speaker Diarization datathon; runner-up, Bengali Olympiad Math Question Solving with LLMs.